Embedded System Lab Manual

Subject Name: Embedded System Lab

Subject Code: 7IT06

Experiment No.: 4

Title: Write and upload a basic C language code on an Arduino Uno board for interfacing Push Button with LED.

Objectives:

1. To understand digital input and output interfacing with Arduino Uno.
2. To interface a push button as input and control an LED as output.
3. To develop and upload a basic C program to demonstrate switch-controlled LED operation.

Required Components/Kit:

- Arduino Uno Board
- USB Cable
- Computer with Arduino IDE Installed
- 1 x LED
- 1 x 220Ω Resistor
- Push Button
- Breadboard and Jumper Wires

Theory:

Arduino Uno is based on the ATmega328P microcontroller and provides multiple digital I/O pins. A push button is a digital input device that provides either HIGH (1) or LOW (0) logic when pressed or released. An LED is a digital output device that can be switched ON or OFF. To avoid floating input, a pull-down resistor is used with the push button, ensuring the pin reads LOW when the button is not pressed.

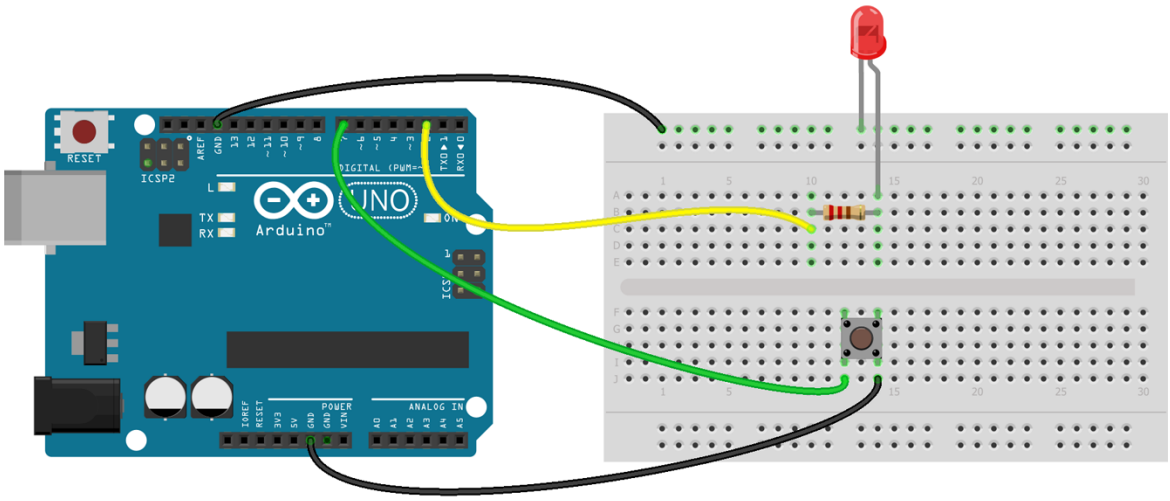
Procedure:

1. **Connect the Arduino Uno** to the computer using a USB cable.

2. Assemble the **circuit** on breadboard as per diagram.
3. **Open Arduino IDE** on your computer.
4. **Write the following C code** in the IDE.
5. Click **"Verify"** (✓ icon) to compile the code.
6. Select the **correct COM port** and **board (Arduino Uno)** from Tools menu.
7. Click **"Upload"** (→ icon) to upload the code to the board.
8. **Press and release the button** → observe LED turning ON/OFF.

Circuit Diagram (Description):

1. Connect LED anode to Arduino digital pin 13 via 220Ω resistor; cathode to GND.
2. Connect one terminal of push button to 5V.
3. Connect the other terminal of button to Arduino pin 2.
4. Also connect a 10kΩ resistor between button pin and GND (pull-down).



Code:

```
int buttonPin = 2;                    // Push button connected to pin 2
int ledPin = 13;                      // LED connected to pin 13
int buttonState = 0;                  // Variable to store button status
void setup()
{
        pinMode(buttonPin, INPUT);    // Set pin 2 as input
        pinMode(ledPin, OUTPUT);      // Set pin 13 as output
}
void loop()
```

```
{
    buttonState = digitalRead(buttonPin);          // Read button state
    if (buttonState == HIGH)
    {
        digitalWrite(ledPin, HIGH);                // LED ON
    }
    else
    {
        digitalWrite(ledPin, LOW);                 // LED OFF
    }
}
```

Working Principle:

- When the button is pressed, the input pin reads HIGH, and the program turns the LED ON.
- When the button is released, the input reads LOW, and the program turns the LED OFF.

Observation Table:

Sr. No	Button State	LED State
1	Not Pressed	OFF
2	Press	ON

Result:

The push button was successfully interfaced with the LED using Arduino Uno. The LED switched ON when the button was pressed and turned OFF when released.

Conclusion:

This experiment demonstrated the interfacing and controlling of an output device (LED) using a push button.